
NeuroCLIP + LATA: Action Semantics and Latency-Aware Alignment for EEG-to-Video Decoding

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Abstract

Decoding natural video perception from non-invasive electroencephalography (EEG) is an important step toward practical brain–computer interfaces, but current EEG-to-video benchmarks leave two questions underexplored: which visual concepts are most reliably represented in EEG, and when does the corresponding neural evidence emerge after stimulus onset? We study these questions on SEED-DV, a 40-concept natural video EEG benchmark, through two complementary contributions. First, we introduce **NeuroCLIP**, a CLIP-aligned EEG retrieval probe that maps EEG into a frozen semantic vision-language space and evaluates concept-level decodability through 40-way retrieval. NeuroCLIP achieves 4.60% Recall@1 and 17.47% Recall@5 on concept retrieval, exceeding chance (2.5% / 12.5%) and matching the scale of supervised EEG2Video-style classification baselines (4.37% top-1, 17.17% top-5). More importantly, it turns aggregate performance into an error-analysis tool: CLIP embedding isolation does not predict EEG decodability ($r=0.036$, $p=0.827$), while action-rich concepts such as sports, music, and people are substantially more decodable than passive scenes (6.8% vs. 3.8%, $p<10^{-7}$). Second, we propose **LATA**, a latency-aware temporal alignment module that learns a soft distribution over biological neural delays during EEG–video contrastive training. On all 20 SEED-DV subjects, LATA recovers a consistent nonzero delay preference: 14/20 subjects peak at a two-chunk lag, 6/20 peak at a one-chunk lag, and no subject peaks at zero lag, with mean expected delay 1.58 ± 0.05 chunks (≈ 790 ms). These results indicate that progress on EEG-to-video decoding requires attention to both stimulus semantics and neural response latency, not only decoder architecture.

1 Introduction

Decoding visual experience from brain activity is a long-standing goal in multimodal AI and brain–computer interfaces (BCIs). If a model can infer what a person is seeing from neural activity alone, it can support assistive communication, closed-loop rehabilitation, and scientific analysis of visual processing. Functional MRI has enabled impressive static image and video reconstruction because it captures spatially detailed cortical activity, but it is expensive, immobile, and temporally slow [Takagi and Nishimoto, 2023, Chen et al., 2023]. Electroencephalography (EEG), by contrast, is non-invasive, portable, and temporally precise, making it a realistic sensing modality for deployable BCIs. The difficulty is that EEG is noisy, subject-specific, spatially coarse, and delayed relative to the stimulus that caused it.

Recent work shows that EEG contains decodable visual information. EEG2Video introduced the SEED-DV dataset and demonstrated that 62-channel EEG can support concept, color, and motion decoding from short natural video clips [Liu et al., 2024]. Following work has used diffusion models, video priors, and multimodal embedding spaces to improve brain-to-vision generation [Li et al., 2024, Liu et al., 2025, Zhou et al., 2026]. However, a single aggregate classification score does not reveal what the EEG signal actually represents. A model may do well on some concepts and fail on others

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Brain-to-Video Decoding from EEG: Motivation & Gap

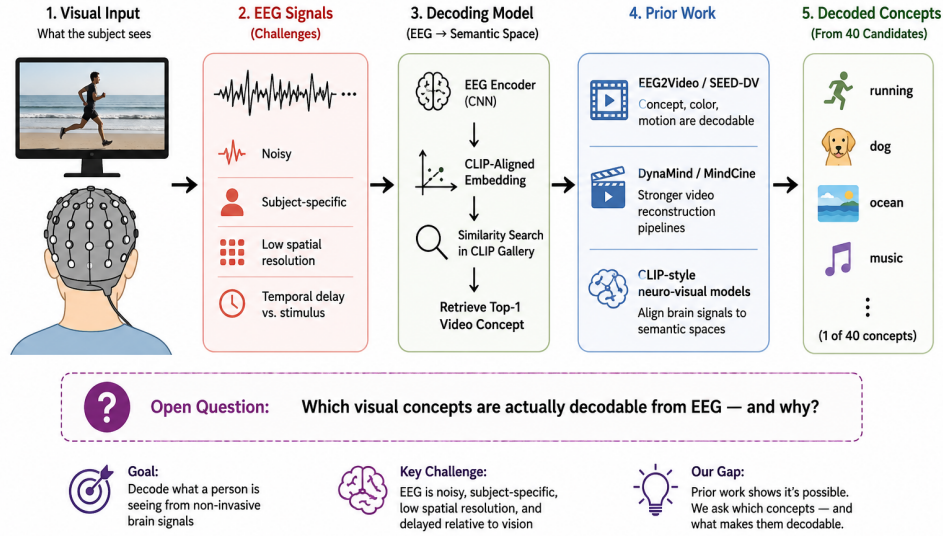


Figure 1: Brain-to-Video Decoding from EEG: Motivation and Gap. A subject views a video clip, 62-channel EEG is encoded into a CLIP-aligned embedding, and the closest concept is retrieved from 40 candidates. Prior work establishes that decoding is possible but reports only aggregate scores. We ask which concepts are decodable and why — and when the neural evidence arrives.

for reasons that have little to do with the visual embedding geometry used by the decoder. In addition, most EEG-video alignment methods treat EEG and video as synchronous streams. This assumption is biologically wrong: visual responses arrive after a latency, with early sensory components around 100 ms and later semantic components several hundred milliseconds after stimulus onset [Luck, 2014].

In this paper, we address both questions through two contributions. NeuroCLIP is a lightweight CLIP-aligned EEG retrieval probe that maps EEG features into a frozen CLIP concept gallery and evaluates per-concept retrieval, turning CLIP from a generative backbone into an analysis instrument that tests whether decodability follows CLIP geometry, semantic category, or protocol artifacts. LATA (Latency-Aware Temporal Alignment) is a differentiable alignment module that learns a soft distribution over candidate neural delays during EEG-video contrastive training, recovering response latency without supervision. Together, NeuroCLIP characterizes which concepts are represented in EEG, and LATA estimates when the corresponding neural evidence arrives. We evaluate both on SEED-DV under leave-one-block-out within-subject cross-validation, with audits for normalization leakage and run-position artifacts.

2 Related Work

Brain-to-vision decoding. Early visual decoding work focused on fMRI because of its high spatial resolution. Recent systems reconstruct images from fMRI using latent diffusion priors [Takagi and Nishimoto, 2023, Ozcelik and VanRullen, 2023] and extend reconstruction to naturalistic video [Chen et al., 2023]. EEG-based visual decoding is more challenging, but its temporal resolution and portability are valuable. Li et al. [Li et al., 2024] decode visual experience from EEG using guided diffusion, while EEG2Video introduces SEED-DV and shows that concept, color, and motion attributes are decodable from 62-channel EEG [Liu et al., 2024]. DynaMind [Liu et al., 2025] and MindCine [Zhou et al., 2026] further study dynamic visual decoding, temporal consistency, and data efficiency. Our work differs by emphasizing semantic and temporal diagnostics rather than photorealistic reconstruction quality alone.

Contrastive multimodal alignment. Contrastive learning has become a standard way to align modalities into shared embedding spaces. CLIP aligns image and text representations using large-scale contrastive supervision [Radford et al., 2021]; VideoCLIP and related models extend this idea to

video-language representations [Xu et al., 2021]. Neuroscience-inspired systems have used CLIP-like spaces to map neural signals onto visual or linguistic latents [Scotti et al., 2024, Fei et al., 2024]. Frozen semantic spaces provide a common coordinate system for retrieval, generation, and analysis. NeuroCLIP follows this direction, but uses the frozen CLIP space mainly as a concept-level probe: instead of asking only whether EEG can retrieve the correct concept, we ask what predicts success or failure across the 40 SEED-DV concepts.

Temporal neural alignment. Temporal alignment is a classic problem in multimodal learning. Dynamic Time Warping aligns two sequences under monotonic timing constraints [Sakoe and Chiba, 1978], and cross-attention learns soft correspondences between sequence elements in transformer-style models [Vaswani et al., 2017]. VideoMAE and related self-supervised video encoders learn useful spatiotemporal features from masked video prediction [Tong et al., 2022]. In EEG, however, temporal alignment must also respect biological latency: the response to a visual event is observed after the event, not at the same timestamp. Event-related potential studies identify a broad hierarchy of visual responses: early sensory components (P100, ~ 100 ms), mid-latency attention components (N200/P300, 200–400 ms), and late semantic components (N400, late positive complex, 400–1000 ms) [Luck, 2014, Kutas and Federmeier, 2011]. LATA builds this latency range into the alignment module by learning a soft distribution over past video chunks spanning up to $\Delta \times 500$ ms.

Protocol artifacts and EEG benchmark validity. Li et al. [Li et al., 2021] demonstrated that block designs cause classifiers to learn temporal drift rather than stimulus-evoked activity, and Ahmed et al. [Ahmed et al., 2021] argue that randomization and careful splitting are essential for object decoding. We address these concerns on SEED-DV through train-only normalization, leave-one-block-out evaluation, and run-position stratification before evaluating NeuroCLIP and LATA.

Positioning. To our knowledge, this work is the first to use a CLIP-aligned EEG retrieval model primarily as a concept-level *decodability probe* on SEED-DV, rather than only as a latent conditioning mechanism for generation. It also introduces an explicit learned EEG–video delay distribution for SEED-DV and reports the resulting population-level latency profile. Our emphasis is diagnostic: we ask which visual semantics EEG contains and when those semantics appear in the neural stream.

3 Problem Statement

Let $s \in \{1, \dots, S\}$ index subjects, $b \in \{1, \dots, 7\}$ session blocks, $c \in \{1, \dots, 40\}$ concept labels, and $m \in \{1, \dots, 5\}$ clip position within a concept run. Each EEG trial is $\mathbf{X}^{(s,b,c,m)} \in \mathbb{R}^{C \times T}$ with $C=62$ channels. For raw EEG, $T=400$ timepoints at 200 Hz; for differential entropy (DE) or power spectral density (PSD), $T=5$ frequency bands. The paired video clip is $\mathbf{Y}^{(b,c,m)}$, and the semantic label is $y^{(s,b,c,m)} = c$.

The supervised classification baseline learns an EEG encoder $f_\theta : \mathbb{R}^{C \times T} \rightarrow \mathbb{R}^d$ and classifier $g_\phi : \mathbb{R}^d \rightarrow \Delta^{40}$ by minimizing

$$\mathcal{L}_{\text{cls}} = -\frac{1}{N} \sum_{i=1}^N \log [g_\phi(f_\theta(\mathbf{X}_i))]_{y_i}. \quad (1)$$

NeuroCLIP instead maps EEG into a frozen CLIP gallery $\{\mathbf{z}_1, \dots, \mathbf{z}_{40}\}$ and evaluates retrieval by cosine similarity. LATA partitions EEG and video into K temporal chunks and learns a delay distribution over $\delta \in \{0, \dots, \Delta\}$ so that EEG chunk k is aligned to stimulus evidence from earlier video chunks.

4 Proposed Method

4.1 NeuroCLIP: A CLIP-Aligned Concept Probe

NeuroCLIP is designed to evaluate which visual concepts are recoverable from EEG and which stimulus properties predict that recoverability. Given an EEG trial $\mathbf{X}_i$, an EEG encoder h_θ produces a normalized embedding $\mathbf{e}_i = h_\theta(\mathbf{X}_i) / \|h_\theta(\mathbf{X}_i)\|_2$. We compare this embedding against a frozen CLIP gallery. For concept-level retrieval, each concept c has a frozen target vector $\mathbf{z}_c$ derived from

CLIP text and image embeddings; our best condition uses an average of text and image concept features. Similarity is

$$q_\theta(c \mid \mathbf{X}_i) = \frac{\exp(\text{sim}(\mathbf{e}_i, \mathbf{z}_c)/\tau)}{\sum_{c'=1}^{40} \exp(\text{sim}(\mathbf{e}_i, \mathbf{z}_{c'})/\tau)}, \quad (2)$$

where sim is cosine similarity and τ is a temperature. The training objective is the 40-way InfoNCE loss:

$$\mathcal{L}_{\text{NCLIP}} = -\frac{1}{N} \sum_{i=1}^N \log q_\theta(y_i \mid \mathbf{X}_i). \quad (3)$$

At inference, the predicted concept is the nearest CLIP gallery item: $\hat{y}_i = \arg \max_c \text{sim}(\mathbf{e}_i, \mathbf{z}_c)$.

The EEG encoder is intentionally lightweight to reduce the extent to which results are driven by decoder capacity. A pointwise 1×1 convolution learns spatial electrode mixing from the 62 EEG channels. For raw EEG, three temporal convolution blocks downsample the 400 timepoints before adaptive average pooling; for DE/PSD features, two smaller temporal convolution blocks operate over the five frequency-band positions. A dropout layer and linear projection map the pooled representation to the CLIP embedding dimension, followed by ℓ_2 normalization. For chunked variants, the same encoder is shared across K chunks and a learned scalar attention score pools chunk embeddings into a single trial representation.

The target gallery is frozen throughout training. We evaluate image-only CLIP targets, text-only targets, and a text+image average; the best final condition uses the averaged target because it combines visual exemplars with stable semantic labels. Importantly, NeuroCLIP avoids within-concept false negatives: all clips with the same concept are assigned the same concept-level positive embedding, and the negatives are the remaining 39 concepts. This makes the objective better matched to the analysis than clip-instance contrastive learning, because SEED-DV contains multiple clips per category while the primary label in our evaluation is concept identity.

This formulation gives the same headline metric as classification (40-way Recall@1), but it also supports concept-level analyses. We compute per-concept retrieval accuracy, CLIP isolation (one minus mean cosine similarity to all other concept embeddings), CLIP consistency across clip exemplars, semantic category membership, and EEG representational similarity. If CLIP geometry were the main reason some classes were easier, isolated CLIP concepts should have higher EEG retrieval accuracy. If brain semantics matter, then concept categories with stronger perceptual or action content should dominate.

4.2 LATA: Latency-Aware Temporal Alignment

NeuroCLIP is concept-level: it asks what semantic information survives in EEG. LATA addresses the complementary temporal question. A 2-second EEG clip and its paired video are partitioned into K chunks: $\mathbf{E} = [\mathbf{e}_1, \dots, \mathbf{e}_K] \in \mathbb{R}^{K \times d}$ and $\mathbf{V} = [\mathbf{v}_1, \dots, \mathbf{v}_K] \in \mathbb{R}^{K \times d}$. Because neural responses lag the stimulus, chunk $\mathbf{e}_k$ may reflect video evidence from $\mathbf{v}_{k-\delta}$ rather than $\mathbf{v}_k$. LATA therefore maintains trainable logits $\ell \in \mathbb{R}^{\Delta+1}$ over candidate delays:

$$\mathbf{w} = \text{softmax}(\ell), \quad \sum_{\delta=0}^{\Delta} w_\delta = 1. \quad (4)$$

The latency-corrected video representation at chunk k is

$$\tilde{\mathbf{v}}_k = \sum_{\delta=0}^{\Delta} w_\delta \mathbf{v}_{k-\delta}, \quad \mathbf{v}_{k-\delta} = \mathbf{0} \text{ if } k < \delta. \quad (5)$$

The corrected video sequence $\tilde{\mathbf{V}}$ is then used as keys and values in cross-attention with EEG queries:

$$\text{LATA}(\mathbf{E}, \mathbf{V}) = \text{MultiHeadAttn}(\mathbf{Q} = W_q \mathbf{E}, \mathbf{K} = W_k \tilde{\mathbf{V}}, \mathbf{V} = W_v \tilde{\mathbf{V}}). \quad (6)$$

We optimize the EEG and video chunk embeddings with an InfoNCE objective:

$$\mathcal{L}_{\text{LATA}} = -\frac{1}{BK} \sum_{i=1}^{BK} \log \frac{\exp(\text{sim}(\mathbf{n}_i, \tilde{\mathbf{v}}_i)/\tau)}{\sum_{j=1}^{BK} \exp(\text{sim}(\mathbf{n}_i, \tilde{\mathbf{v}}_j)/\tau)}. \quad (7)$$

Gradients flow through $\mathbf{w} = \text{softmax}(\ell)$, so the delay distribution is learned without any explicit delay labels. Here index i ranges over all BK chunk instances in the batch; the positive pair for EEG chunk k of clip n is the latency-corrected video chunk k of the same clip n , and all other $BK-1$ chunk instances serve as negatives. After training, $\mathbb{E}[\delta] = \sum_{\delta} \delta w_{\delta}$ gives an interpretable estimate of neural response delay in chunk units.

LATA is deliberately parameter-efficient: the only latency-specific parameter is the vector ℓ , so any learned nonzero lag cannot be attributed to a large delay-prediction subnetwork. The module can be inserted anywhere a standard cross-attention layer would align paired temporal streams. In our experiments, EEG chunks are encoded with a shallow temporal-spatial CNN, and video chunks are represented by frozen CLIP frame embeddings sampled at matched chunk timestamps. During training, LATA uses contrastive chunk matching to push each neural chunk toward the latency-corrected visual chunk at the same index and away from other chunks in the batch. During analysis, we inspect both the peak delay $\delta^* = \arg \max_{\delta} w_{\delta}$ and the expected delay $\mathbb{E}[\delta]$.

5 Experimental Methodology

Dataset. We use SEED-DV [Liu et al., 2024], which contains 62-channel EEG recorded at 200 Hz while subjects watched natural 2-second video clips. The benchmark contains 40 concept categories, 5 clips per concept per block, and 7 session blocks per subject. We evaluate raw EEG (62×400), DE features, and PSD features. DE and PSD compress each trial into 5 canonical frequency bands: δ , θ , α , β , and γ .

Splits and preprocessing. For supervised baselines and NeuroCLIP, we use within-subject leave-one-block-out cross-validation: six blocks for training and one held-out block for testing, rotated over the seven blocks. This prevents train/test session overlap. We use train-only normalization statistics for leakage-controlled conditions and apply them to validation/test folds. NeuroCLIP is evaluated on 21 subject-recording files from the SEED-DV preprocessing artifacts, corresponding to the 20 subjects plus one repeated subject recording used in the available files. LATA is run on all 20 subjects with six train sessions and one held-out validation session.

Visual targets. For NeuroCLIP, we precompute frozen CLIP ViT-B/32 embeddings for concept labels and/or representative visual frames. Concept-level galleries contain 40 normalized vectors, one per SEED-DV class. For LATA, video features are extracted at the chunk level: each 2-second clip is divided into four 0.5-second chunks, and one visual feature is assigned to each chunk. These frozen features ensure that the trainable parameters are concentrated in the EEG encoder and the alignment module, not in a large visual backbone.

Models and metrics. The supervised baseline uses the EEG2Video GLMNet-style feature classifier for DE/PSD features and a raw-EEG-compatible encoder for raw signals. NeuroCLIP uses the same within-subject fold structure but replaces the classifier with a CLIP retrieval objective. We report Recall@1, Recall@5, Recall@10, mean reciprocal rank (MRR), and per-concept retrieval. LATA uses $K=4$ chunks of 0.5 s each, $d_{\text{model}}=128$, maximum delay $\Delta=3$, batch size 64, AdamW with learning rate 3×10^{-4} , cosine annealing, and temperature $\tau=0.07$. For LATA, the primary result is the learned delay distribution rather than a classification accuracy gain.

Statistical analysis. We compare retrieval scores to chance using subject-level tests and report concept-level analyses across the 40 classes. To test whether visual embedding geometry explains EEG decodability, we correlate per-concept Recall@1 with CLIP isolation and CLIP confusion. To test semantic effects, we compare activity categories (sports, music, people) against passive categories. For protocol validity, we report run-position accuracy profiles, session trends, and within-concept prediction consistency against empirical random baselines.

6 Results and Discussion

6.1 Baseline and NeuroCLIP Retrieval Performance

Table 1 summarizes the main 40-way results. The supervised DE baseline reaches 4.37% top-1 and

17.17% top-5 accuracy, above chance but well below the originally reported EEG2Video GLMNet result. NeuroCLIP reaches 4.60% concept Recall@1 and 17.47% concept Recall@5 using CLIP text+image targets, consistent with the modality comparison in Figure 2A. The absolute gain over the supervised DE classifier is modest; the main benefit of NeuroCLIP is that retrieval exposes concept-level structure that a single aggregate classification score cannot capture.

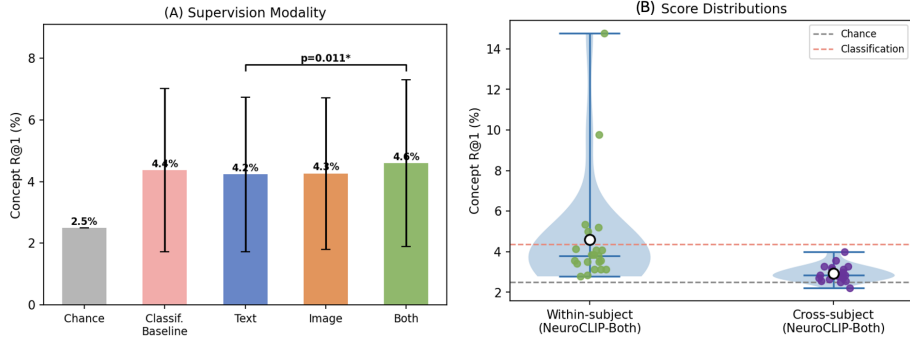


Figure 2: **NeuroCLIP summary diagnostics.** NeuroCLIP summary diagnostics. (A) Concept Recall@1 across supervision conditions compared to the 2.5% chance level; the fused text+image target (Both) gives the best performance at 4.60% ($p=0.011$ vs. chance). (B) Score distributions show high subject variability within-subject and weaker cross-subject generalization, motivating within-subject evaluation. Error bars show ± 1 SD across subject-recordings.

Table 1: Main 40-way concept results. Values are mean $\pm$ standard deviation across subject-recordings for NeuroCLIP and across subjects for supervised baselines. Chance is 2.5% for top-1/R@1 and 12.5% for top-5/R@5.

Condition	Top-1 / R@1 (%)	Top-5 / R@5 (%)
Supervised DE baseline	4.37 ± 2.64	17.17 ± 5.44
Supervised DE, train-only normalization	4.09 ± 2.62	16.84 ± 5.19
Supervised PSD baseline	4.15 ± 2.60	17.15 ± 4.89
Supervised Raw EEG baseline	3.79 ± 1.56	15.79 ± 3.54
NeuroCLIP, DE $\rightarrow$ CLIP image	4.26 ± 2.46	17.51 ± 4.43
NeuroCLIP, DE $\rightarrow$ CLIP text+image	4.60 ± 2.70	17.47 ± 5.14
Chance	2.50	12.50

The ranked retrieval metrics show that the signal is not limited to exact top-1 hits. NeuroCLIP obtains R@1/R@5/R@10/R@20 of 4.57%, 18.29%, 32.98%, and 58.0%, respectively, and MRR of 0.143 versus a chance MRR of 0.107 ($p = 7.35 \times 10^{-5}$). These results indicate statistically reliable signal despite the difficulty of 40-way EEG decoding from short natural videos. Category-level retrieval is also above chance: 15.5% R@1 over 8 semantic categories versus 12.5% chance, and intra-category retrieval reaches 24.0% versus 20.0% chance.

Performance is highly subject dependent, as shown in Figure 2B. Some recordings approach 10–15% R@1, while many are only slightly above chance. This subject variability is a central practical limitation for EEG-based BCIs and explains why we use within-subject models rather than a single pooled cross-subject model as the main evaluation. It also motivates concept-level analysis: when aggregate accuracy is low, a per-class probe can reveal whether the model is failing uniformly or whether a subset of concepts carries most of the decodable signal.

The comparison between DE and raw EEG is also informative. Raw EEG preserves the full 400-sample temporal waveform, but it underperforms the DE baseline in the supervised setting and in NeuroCLIP-style retrieval. This does not imply that temporal information is irrelevant; rather, it indicates that the current raw encoder does not extract robust temporal features from the available amount of subject-specific training data. The compressed DE representation may discard within-clip dynamics, but it also averages over high-frequency noise and reduces the dimensionality of the learning problem. This tradeoff motivates the two directions we pursue: use DE features for stable concept-level semantic analysis, and use LATA on chunked representations to study temporal alignment separately from aggregate concept classification.

6.2 Validity Checks: Leakage and Sequence Shortcuts

Because SEED-DV uses concept-blocked presentation, we first checked whether retrieval could be explained by simple protocol artifacts. Train-only normalization did not produce a large collapse: DE top-1 changed from 4.37% to 4.09%, while PSD changed from 4.15% to 4.29%. This indicates that global normalization was not the main source of above-chance performance, although fold-level variance was affected and train-only normalization should be used for future ablations.

We also stratified predictions by clip position within the 5-clip same-concept run. DE accuracy across positions 1–5 was [4.40, 4.29, 4.48, 4.38, 4.10]%, and PSD was [3.84, 4.39, 4.29, 4.13, 4.31]%. Neither profile shows a monotonic run-position trend. Within-concept consistency is above random (DE: 24.3% vs. 17.8%), indicating that predictions are semantically structured, while the flat run-position profile provides no evidence for a simple adaptation or anticipation shortcut. These audits do not rule out every possible block-design artifact, but they reduce the likelihood that the main results are driven by the most direct sequence-position confound.

The remaining risk is block-level semantic priming: a subject sees five clips from the same concept after a cue, so the EEG may contain concept expectation in addition to stimulus-evoked visual processing. Our analyses partially address this by using held-out session blocks and by showing no monotonic accuracy profile across positions 1–5, but they cannot fully separate perceptual evidence from concept-set maintenance. This distinction matters for deployment: a BCI user would not normally experience a repeated concept cue immediately before each decoded event. For this reason, we treat SEED-DV as a useful but partly structured benchmark, and we emphasize relative comparisons and error analysis over absolute claims about open-world decoding accuracy.

6.3 What Predicts EEG Decodability?

The most natural hypothesis is that EEG retrieval succeeds for concepts that are well separated in CLIP space. Figure 3A shows that this is not what we observe: per-concept CLIP isolation is uncorrelated with NeuroCLIP Recall@1 ($r = 0.036$, $p = 0.827$), and CLIP confusion is also uncorrelated with EEG confusion (Spearman $\rho = -0.024$, $p = 0.509$). In other words, NeuroCLIP is not merely easier when the frozen gallery is geometrically convenient.

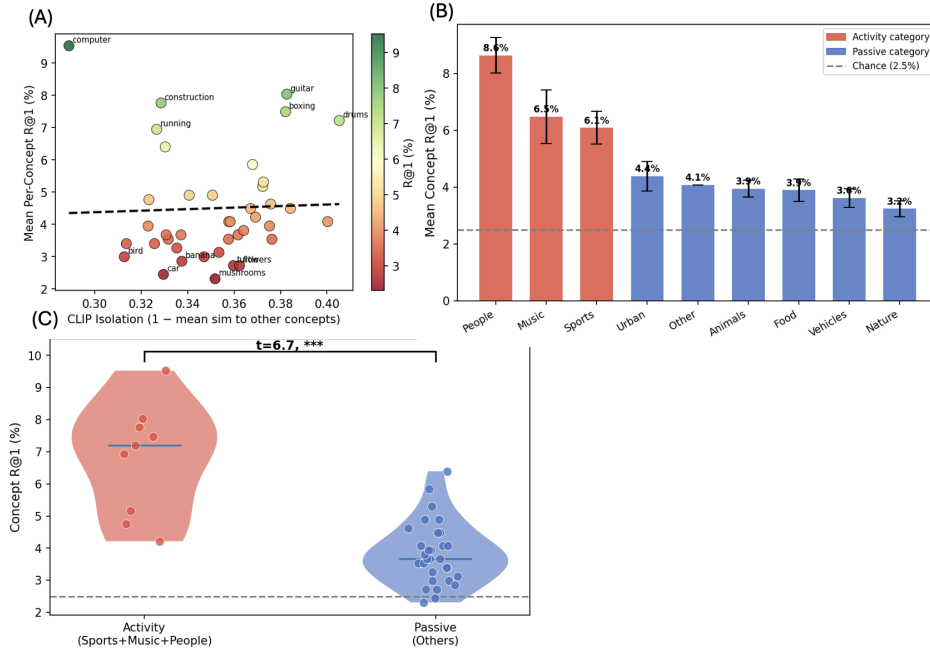


Figure 3: **NeuroCLIP concept-level error analysis.** (A) CLIP geometry does not predict EEG decodability: per-concept CLIP isolation is uncorrelated with Recall@1 ($r = 0.036$, $p = 0.827$). (B) Decodability is structured by semantic content: activity categories (people, music, sports) reach 6.1–8.6% R@1, while passive categories (urban, animals, food, vehicles, nature) cluster near 4%. (C) Aggregating across the activity/passive split confirms the effect (6.79% vs. 3.83%, $t = 6.72$, $p = 5.95 \times 10^{-8}$).

The semantic category effect shown in Figure 3B–C explains why aggregate accuracy is low but not random: much of the recoverable signal is concentrated in a subset of concepts. If evaluation is restricted to the top 10 most decodable concepts, R@1 rises to 6.97%, compared with 4.50% over all 40 concepts. Per-band DE comparisons show no significant activity–passive difference across delta, theta, alpha, beta, or gamma bands, suggesting that the advantage is not simply a global EEG power effect.

The effect is also not a clean binary. Some non-action concepts, including “computer” and “construction,” rank highly, while some dynamic animal or vehicle concepts rank lower. This points to a mixture of action, salience, social content, object typicality, and within-concept visual consistency. A useful follow-up would annotate each concept along these axes and model per-concept EEG decodability as a multivariate response.

6.4 LATA Learns a Consistent Nonzero Neural Lag

LATA was first validated on paired synthetic sequences with known ground-truth delays $\delta_{\text{true}} \in \{0, 1, 2, 3\}$, then applied to all 20 SEED-DV subjects.

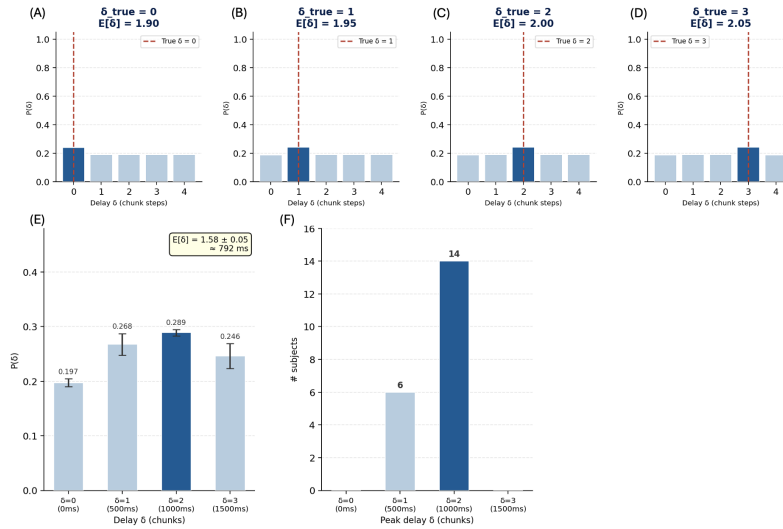


Figure 4: **LATA delay-learning results.** (A–D) Synthetic validation: when neural activity is generated from a stimulus shifted by a known delay $\delta_{\text{true}} \in \{0, 1, 2, 3\}$, LATA’s learned distribution peaks at the correct value in all four cases. (E) On SEED-DV across 20 subjects, the population mean delay distribution $[0.197, 0.268, 0.289, 0.246]$ concentrates on nonzero lags, with expected delay 1.58 ± 0.05 (≈ 790 ms). (F) Peak delay per subject: 14/20 subjects peak at $\delta^* = 2$ (1000 ms per-chunk lag), 6/20 at $\delta^* = 1$ (500 ms), and none at $\delta^* = 0$. The population mean expected delay $E[\delta] \approx 790$ ms reflects the soft distribution rather than the argmax, supporting latency-aware rather than synchronous EEG–video alignment.

On real EEG, LATA provides stronger evidence for interpretable latency estimation than for improved held-out retrieval. The learned nonzero delay is later than early visual components and is more consistent with semantic or late positive processing, but $K = 4$ chunks gives only coarse 500 ms resolution. The $\delta = 2$ peak should therefore be read as evidence for delayed stimulus evidence rather than a single discrete ERP component; the 1000 ms argmax lag falls at the late end of established semantic ERP components (P300/N400, late positive complex), and the coarse chunk resolution means the true neural peak could lie anywhere in the 500–1000 ms window. The population distribution $\bar{w} = [0.197, 0.268, 0.289, 0.246]$ is notably diffuse — the peak weight ($w_2=0.289$) exceeds the trough ($w_0=0.197$) by only 0.09 — contrasting with the sharply peaked distributions recovered in synthetic validation (Figure 4A–D). This softness likely reflects the coarse 500 ms chunk resolution and noisy frozen CLIP frame features, and may also indicate that late ERP energy is genuinely distributed across the 500–1000 ms window rather than concentrated at a single lag.

Thus, LATA is best understood as an alignment diagnostic rather than a finished retrieval model. It does not yet solve chunk-level EEG–video retrieval, but it provides evidence against the zero-lag assumption used by many simple alignment pipelines. The remaining softness of the learned distribution may reflect coarse chunking, noise in CLIP frame features, or semantic processing distributed across multiple late ERP components.

The latency result also clarifies why whole-clip concept retrieval and chunk-level temporal alignment should be evaluated separately. A whole-clip embedding can succeed by averaging delayed neural evidence over the full 2-second window; it does not need to know which video moment produced which EEG response. LATA, by contrast, explicitly tests whether the model can align local neural and visual events. Its current behavior indicates that coarse temporal delay is learnable, but that fine-grained chunk retrieval remains substantially harder than concept retrieval. This gap is an important negative result: dynamic tracking cannot be assumed from above-chance whole-clip classification alone.

7 Conclusion and Future Work

This project studied EEG-to-video decoding through the lens of semantic and temporal interpretability. NeuroCLIP showed that 40-way EEG concept retrieval is above chance and comparable in scale to supervised EEG2Video-style classification, but its main value is diagnostic: CLIP geometry does not explain which concepts are decodable, while action-rich semantic content does. LATA addressed the temporal side of the same problem by learning neural response delay end-to-end. Across 20 subjects, the learned distributions consistently avoid zero lag and concentrate around one to two 0.5-second chunks.

Together, these findings refine how SEED-DV results should be interpreted. Above-chance concept accuracy is not uniformly distributed across the stimulus set, nor is EEG evidence temporally synchronous with the video stream. The benchmark contains meaningful neural signal, but that signal is selective and delayed. A model that improves average accuracy without examining these two factors may obscure the underlying structure of the decoding problem. Diagnostic tools such as NeuroCLIP and LATA instead identify where the signal is strongest and which alignment assumptions are least justified.

Several limitations remain. Absolute EEG decoding accuracy is low, and the largest NeuroCLIP gains are explanatory rather than performance-improving. The SEED-DV blocked design may contain subtler concept-priming or session-drift effects beyond what run-position stratification captures, and because SEED-DV concepts differ along correlated visual dimensions (motion, faces, object scale), the action advantage cannot yet be cleanly attributed to action semantics alone. LATA uses coarse 500 ms chunks and frozen CLIP frame features rather than spatiotemporal video encoders, so its latency estimates are approximate. Most importantly, both analyses are still within-dataset: they show structure in SEED-DV but do not yet prove that the same semantic and temporal patterns generalize.

Future work should evaluate NeuroCLIP on held-out concepts and additional EEG datasets such as THINGS-EEG, where stimulus diversity is larger and block effects can be tested more aggressively. For LATA, the next step is finer temporal resolution (e.g., 100 ms chunks), stronger frozen video encoders such as VideoMAE, and a direct comparison between zero-lag cross-attention, fixed-lag alignment, and learned-lag alignment. More broadly, our results support reporting concept-level decodability, protocol audits, and learned temporal alignment diagnostics alongside aggregate accuracy in EEG-to-video benchmarks.

The most important modeling extension is to unify the two diagnostics rather than treating them as separate modules. A future *Latency-Aware NeuroCLIP* model would use LATA to compute latency-corrected EEG chunk embeddings, then map those temporally aligned embeddings into the same CLIP concept space used by NeuroCLIP. The unified model would test whether action-rich concepts remain decodable with latency-corrected chunk embeddings, whether semantic groups prefer different delays, and whether genuine neural-video correspondence can be distinguished from static category matching via ablations against zero-lag and shuffled-chunk baselines.

There are also smaller modeling extensions. NeuroCLIP could be trained with a hierarchical objective that predicts coarse semantic category and fine concept jointly, reducing the penalty for visually plausible confusions. LATA could be made subject-adaptive by learning a population delay prior plus a small subject-specific offset.

A final evaluation priority is moving beyond within-dataset validation. A stronger benchmark would train on one subset of concepts and evaluate on held-out concepts with matched semantic categories, or train on SEED-DV and evaluate on a separate EEG vision dataset with different acquisition conditions. Such tests would separate dataset-specific regularities from reusable neural-visual correspondences.

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